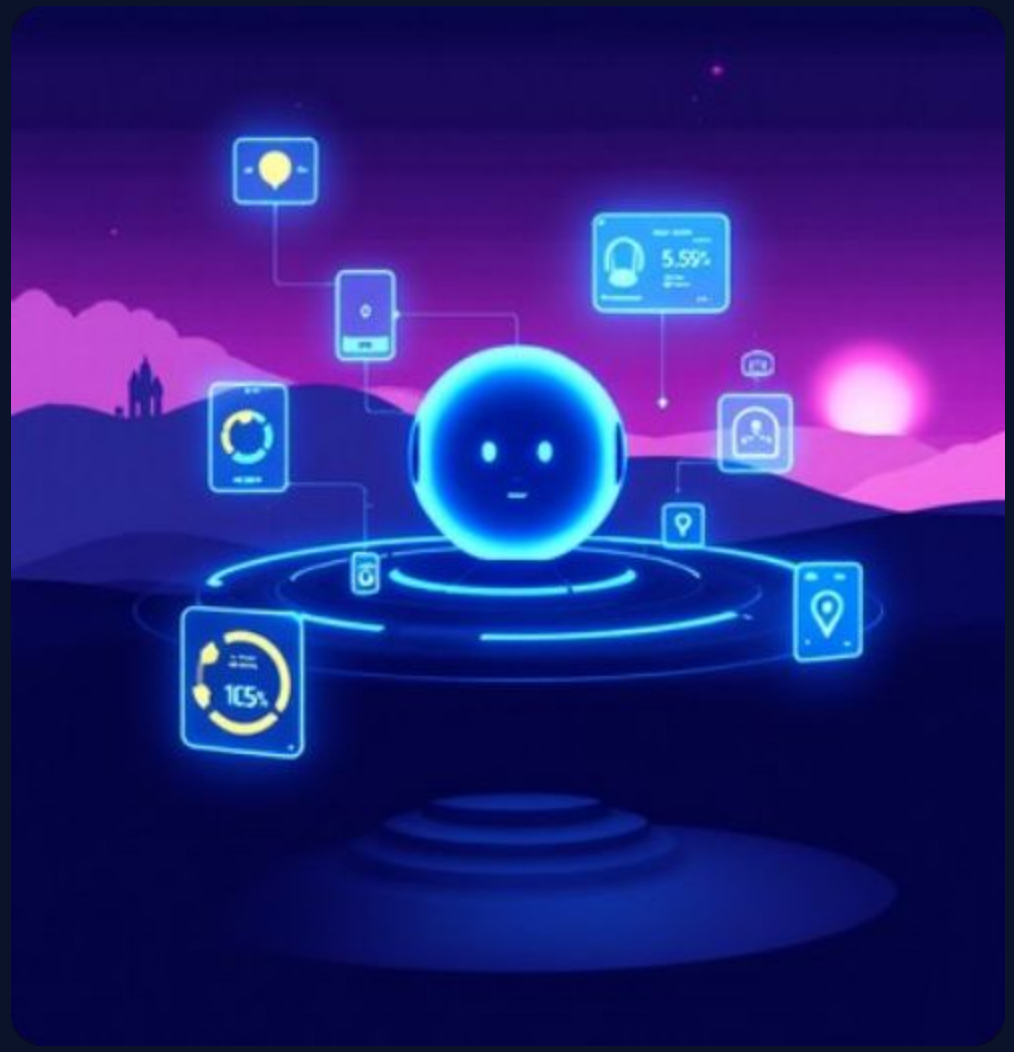


Course: Introduction to Artificial Intelligence

Everyday AI Tools

Session 5 | Professional Prompt Engineering & Mastering Smart Assistants

Instructor & Producer: Amin Maghrebi

Previous Recap & Today's Objectives

Recap of Session 4: The Generative AI Revolution

Understanding LLMs (next-word prediction) and foundational overview of text and image models.

Practical Objectives of This Session:

Prompt Engineering Formula: Learn the principles of writing precise and professional prompts

Mastering Text Assistants: Practical applications of ChatGPT and Claude in work and education

Multimedia AI Tools: Top platforms for generating images, videos, and voices

Automation & Tool Workflows: Automating repetitive tasks to save valuable time

Why Everyday AI Tools Matter?



1. Dramatic Time Savings

Hours of drafting, translation, and summarization in minutes

2. Overcoming the Blank Page Syndrome

Instant brainstorms that end 'Where do I begin?' hesitation

3. 24/7 Specialized Assistant

An always-on tutor, editor, and designer right in your pocket

Key 21st-Century Skill

What is Prompt Engineering?

The Art of Clear Communication with AI

A prompt is any text, question, or instruction given to an AI model to produce a desired output.

Golden Rule: Precise & clear input = Professional & actionable output

Amateur vs. Pro: AI isn't a mind reader; it requires explicit guidance and context

No Coding Required: Mastering AI simply through structured, natural language

4-Part Framework

Principles of Effective Prompt Engineering

1. Define the Role:

Specify the persona or expertise (e.g., 'Act as an experienced physics teacher')

2. Provide Context:

Describe background & target audience (e.g., 'Explain for 12-year-old middle school students')

3. State the Specific Task:

Clearly articulate the exact request without vagueness or ambiguity

4. Specify Output Format & Tone:

Set format & tone (e.g., 'Present as a 3-column table in an engaging, encouraging tone')

Practical Comparison

Real-World Example: Weak vs. Strong Prompt

Weak & Vague Prompt:

"Write an article about exercising."

Result: A generic, repetitive article with no actionable takeaways.

Strong & Professional Prompt:

"Act as a fitness coach. Design a 15-minute home workout routine requiring zero equipment for desk-bound office workers. Present it as a weekly schedule table in an encouraging, motivational tone."

Result: A tailored, structured routine ready to execute immediately!

Versatile Assistant

Practical ChatGPT Use Cases in Work & Study

1. Personal Tutor & Study Partner

Explaining difficult concepts simply and generating practice quizzes directly from study notes

2. Business Writing & Resume Polishing

Drafting formal business emails, optimizing CVs/resumes, and conducting mock job interviews

3. Excel Formulas & Spreadsheet Analysis

Formulating complex Excel functions with clear explanations and analyzing statistical datasets

Writing & Analysis Specialist

Leveraging Claude for Work & Creativity

1. Highly Fluent & Nuanced Writing

Composing articles, stories, and copy with an authentic, human tone free of mechanical phrasing

2. Deep Document Synthesis & Long PDFs

Ingesting hundred-page books or contracts to answer nuanced questions with pinpoint accuracy

3. Interactive Visuals & Frontend Code

Transforming text outlines into flowcharts, diagrams, and live previewable interactive UI

Head-to-Head: When to Use ChatGPT vs. Claude?

Everyday Use Case	Top Pick	Key Advantage
Web Search & Voice Conversation	ChatGPT	Live browsing integration & advanced real-time voice mode
Long-Form Writing & Copy Editing	Claude	Superior literary style, natural phrasing & organic prose
Complex Doc Analysis & Coding	Claude	Massive context window & higher syntactic precision in code
Integrated Text & Image Creation	ChatGPT	Built-in DALL-E / image generation directly inside chat

Smart Creative Studio

Practical AI Image Generation & Editing Tools



1. Midjourney & Google ImageFX

Photorealistic imagery, concept art, and posters from text prompts

2. Canva AI Magic Studio

One-click background removal, object erasing, and magic graphic edits

3. Leonardo AI

Generous daily free tier for custom character, game, and logo design

Multimedia Creation

Practical AI Video & Voice Generation Tools

1. ElevenLabs (Hyper-Realistic Voice)

Lifelike text-to-speech, emotional inflections, and seamless automated multi-language dubbing

2. Runway & Kling AI (Video Generation)

Animating still photos and generating short promotional cinematic video clips from text

3. Suno AI (Music & Song Creation)

Generating full broadcast-quality songs with vocals and instrumental background tracks in seconds

Work Smarter, Not Harder

Automation: Streamlining Repetitive Work with AI



1. Automated Meeting Summarization

Transcribing live meetings to text and instantly distilling assigned action items

2. App Integrations (Make & Zapier)

Routing incoming emails to LLMs to categorize inquiries and draft smart replies

3. Custom AI Assistants (Custom GPTs)

Building specialized bots with tailored instructions to handle daily workflows

Content Creation Pipeline

Tool Orchestration: Combining AI for 10x Results

Step 1: Ideation & Scriptwriting with Claude

Drafting a compelling, high-retention 60-second educational video script

Step 2: Concept Art with Midjourney or ChatGPT

Generating 4 cohesive storyboard frames matched to key narrative scenes

Step 3: Voiceover & Motion with ElevenLabs & Runway

Synthesizing rich voice narration and animating still storyboard artwork

Final Result in Under 30 Minutes:

A polished, end-to-end promotional video produced by chaining 4 AI tools!

Live Interactive Workshop

Hands-on Exercise: Rewrite & Elevate the Prompt!

Your 5-Minute Challenge:

Exercise: Elevate the weak prompt below using the 4-part framework (Role, Context, Task, Format):

Raw Prompt: "Give me a travel itinerary for Shiraz."

Tip: Add duration (days), budget tier, interests (historic vs food), companions, and specify a structured table format!

Key Tips & Common Mistakes to Avoid

1. Settling for the Very First Response

Always iterate: use follow-up prompts to refine, condense, or sharpen initial drafts

2. Blindly Trusting Facts & Stats (Hallucinations)

Always cross-check critical legal, medical, financial, and historical facts with authoritative sources

3. Tool Fatigue & Distraction Overload

Instead of chasing hundreds of apps, deeply master 2 or 3 core tools for your actual workflow

Session 5 Summary

1. **The Power of Prompts:** AI output quality is a direct reflection of prompt clarity and precision.

2. **The 4-Part Framework:** Role, Context, Task, and Format unlock consistently elite responses.

3. **Choose the Right Assistant:** ChatGPT for web & voice, Claude for nuanced writing & deep analysis.

4. **Smart Tool Orchestration:** Chaining text, visual, and voice generators multiplies creative velocity.

Frequently Asked Questions

Common Questions on Everyday AI Tools

Is it better to prompt in English or Persian?

Modern LLMs handle Persian exceptionally well; however, image generators still yield sharper details in English.

Is a paid subscription required to get started?

No, the free tiers of ChatGPT, Claude, and Gemini cover over 90% of daily productivity needs.

How should I save and organize my best prompts?

Build a simple prompt playbook in Notion or Apple Notes to quickly reuse and iterate your top formulas.

Weekly Practical Assignment

Session 5 Assignment: Prompt Engineering in Action!

Weekly Mission: Execute a real project using the 4-part framework

Delegate one time-consuming academic or professional task this week to an AI assistant.

1. **Draft 3 Structured Prompts:** Create 3 full prompts following Role, Context, Task, and Format
2. **Side-by-Side Comparison:** Feed the same prompt to both ChatGPT and Claude and compare outputs
3. **Share Insights:** Share your top performing prompt and output in the course community group

Session 5 Conclusion | Introduction to Artificial Intelligence

Thank You for Your Active Participation!

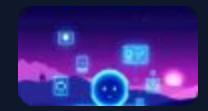
Next Session (Session 6): Ethics, Safety, & the Future of Careers in the AI Era

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